

MAKHACHKALA MEZHDUNARODNYY, Russian Federation

WMO: 377352

Lat: 42.817N

Lon: 47.652E

Elev: 12

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	19.2	11.8	10.0	18.9	15.5	12.1	22.9	33.6	38.5	29.8	36.8	10.1	340	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	91.6	72.7	89.6	72.4	87.7	72.6	77.4	85.4	76.0	84.7	74.8	83.8	12.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	131.8	82.9	73.4	124.3	81.7	71.7	117.1	80.8	40.8	85.5	39.5	83.8	38.2	83.8	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.5	25.0	21.8	8.3	96.7	8.4	1.3	2.3	97.6	-2.7	98.4	-7.4	99.1	-13.5	100.1
			10.5	79.7	5.3	2.9	6.7	81.8	3.6	83.5	0.6	85.1	-3.3	87.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	34.9	35.3	42.1	51.2	63.4	73.4	78.3	77.7	69.3	58.0	46.3	39.8
	DBStd	16.74	7.37	7.67	4.43	6.06	5.56	4.92	3.96	4.37	5.46	6.33	6.78	6.32
	HDD50	1652	469	411	247	52	0	0	0	0	0	14	141	318
	HDD65	4578	934	831	710	414	96	3	0	0	22	227	560	781
	CDD50	3821	0	0	3	89	417	702	877	859	580	262	30	2
	CDD65	1273	0	0	0	1	48	255	412	395	152	10	0	0
	CDH74	11667	0	0	0	14	329	2231	4256	3949	870	18	0	0
	CDH80	4013	0	0	0	0	65	679	1614	1499	156	0	0	0
Wind	WSAvg	9.9	9.8	9.9	10.7	10.0	9.2	9.5	9.7	9.5	10.0	10.1	10.2	10.2
Precipitation	PrecAvg	15.1	1.0	1.1	0.8	0.8	1.5	1.4	1.1	1.2	1.8	1.8	1.4	1.1
	PrecMax	21.4	3.0	4.1	1.9	1.3	2.6	3.1	2.6	3.4	6.5	5.0	4.3	4.1
	PrecMin	11.0	0.2	0.3	0.1	0.2	0.2	0.5	0.4	0.0	0.0	0.3	0.0	0.0
	PrecStd	3.0	0.8	0.9	0.5	0.3	0.7	0.6	0.6	0.9	1.7	1.3	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.8	52.0	60.9	75.6	86.0	91.8	95.1	95.1	87.9	76.6	64.8	55.7
		MCWB	43.7	45.6	51.5	60.9	66.3	71.0	73.7	72.9	71.3	65.1	58.9	49.3
	2%	DB	47.8	48.0	55.4	69.9	81.0	89.3	91.6	91.6	84.1	73.1	60.8	52.2
		MCWB	42.8	43.8	48.2	58.6	66.2	70.7	73.0	73.1	71.5	64.6	56.3	48.0
	5%	DB	44.9	45.0	51.9	64.8	77.3	86.2	89.5	89.6	80.9	70.0	58.6	50.2
		MCWB	41.4	42.1	46.0	55.7	65.0	70.7	72.7	72.9	69.9	63.2	54.3	47.0
	10%	DB	43.0	43.2	49.7	61.2	73.8	84.1	87.6	87.7	78.8	67.9	55.7	48.3
		MCWB	40.4	41.1	45.0	54.5	63.2	69.6	72.7	73.3	69.1	62.0	52.1	45.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	46.8	52.1	62.3	69.4	76.9	78.9	79.4	76.9	67.7	59.9	52.0
		MCDB	49.6	50.8	59.2	73.7	79.0	85.4	86.9	86.4	81.6	73.4	63.4	54.5
	2%	WB	43.4	44.0	49.2	59.1	67.5	74.6	77.3	77.8	73.6	65.6	56.8	49.5
		MCDB	45.5	46.1	54.8	67.9	77.4	83.6	85.9	85.5	80.3	71.0	60.2	51.0
	5%	WB	41.9	42.7	46.7	57.0	66.1	72.8	75.9	76.3	71.8	64.0	54.8	47.7
		MCDB	43.8	44.3	50.6	64.0	75.7	82.5	85.0	84.7	78.9	68.9	58.0	49.6
	10%	WB	40.6	41.3	45.1	54.9	64.6	71.4	74.7	75.1	69.9	62.6	52.8	45.9
		MCDB	42.3	42.7	47.9	60.8	72.8	80.9	84.0	84.1	76.8	67.2	55.7	47.7
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.0	11.7	14.4	16.6	17.2	16.2	16.1	14.5	12.7	11.7	10.3
		MCDBR	14.4	12.6	17.0	19.3	22.0	20.4	19.1	18.9	16.5	15.2	15.8	12.6
		MCWBR	10.7	9.3	11.2	11.3	10.7	10.0	9.3	10.1	9.4	9.7	11.5	9.4
	5% WB	MCDBR	12.2	11.1	14.8	18.1	19.3	17.9	17.0	17.1	15.8	13.7	14.0	10.2
		MCWBR	9.6	8.6	10.5	11.2	10.0	10.0	9.6	10.6	10.0	9.4	10.6	8.3
Clear-Sky Solar Irradiance	taub	0.364	0.401	0.441	0.504	0.478	0.478	0.512	0.511	0.475	0.421	0.380	0.369	
	taud	2.236	2.130	2.075	1.960	2.107	2.165	2.089	2.083	2.140	2.238	2.273	2.231	
	Ebn at Noon	235	245	250	244	255	255	245	239	238	236	228	220	
	Edh at Noon	30	39	46	56	49	47	50	49	43	34	29	28	
All-Sky Solar Radiation	RadAvg	461	710	1032	1444	1761	1955	1924	1726	1325	848	556	412	
	RadStd	54	96	92	149	153	114	141	138	128	103	59	50	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page